

Estimation of electrode porosity from plasma–material interaction measurements

A feasibility study prepared for the stakeholder’s process engineering team

1. Summary and Recommendation

Electrode porosity is presently determined by offline characterization. This assessment considers whether the non-destructive plasma measurements, comprising optical emission spectroscopy, thermal imaging and electrical waveform capture, could estimate it inline instead. On the 20 samples provided the approach is physically sound and merits further investigation, but the dataset is too small to demonstrate a usable measurement. We recommend a proof of concept of 40 to 60 electrodes, and recommend against deploying a model built on the presently supplied data.

Question	Finding
Do the measurements respond to porosity?	Yes. All three measurement types vary with porosity.
Can porosity be estimated at present?	Not to a useful accuracy. On an electrode not previously measured, typical error is 2.85 porosity percentage points, against 3.41 when the average is assumed.*
Is the instrumentation the limiting factor?	No. The limiting factor is the number of electrodes characterized.
What would resolve the question?	A designed study of 40 to 60 electrodes spanning a deliberately wider porosity range.

* Typical error, in porosity percentage points, is the root-mean-square difference between predicted and measured porosity, on the porosity scale itself. An error of 2.85 means predictions are out by roughly 2.85 where the samples supplied span 12.4 points in total. This is not the same as a percentage of the porosity value, which would be a smaller-sounding number.

2. Response of the Measurements to Porosity

Each electrode was measured 100 times, which separates differences between electrode response from other instrumental effects that can give a similar signal. The variation among the electrode samples dominates all three measurement channels and none trends with order of data acquisition, so the relationships reported below are real. Further repeats on these same electrodes would not add new information; additional data must consist of additional electrodes.

A correlation parameter, r , was calculated to describe how closely the measured parameters followed a linear relationship to porosity, along with a 95% confidence interval (95% C.I.). If the study were repeated on different sets of electrodes from the same process, intervals constructed this way would contain the true correlation 95 percent of the time. r ranges between -1 and $+1$, where zero indicates no linear relationship and the sign gives the trend direction. A 95% C.I. excluding zero indicates an established relationship, and its width shows how precisely the value has been determined. For this study, only linear relationships were examined, so a non-linear relationship would not appear. The following table describes the correlations found in the measured data sets and their respective interpretation.

Channel	r	95% CI	Direction and interpretation
Optical emission	+0.52	[+0.10, +0.78]	Higher porosity gives brighter emission at specific wavelengths, while the broad background is unaffected.
Thermal field	+0.41	[−0.04, +0.72]	Higher porosity gives a higher thermal response. Interval includes 0, so no trend established.
Electrical current	−0.58	[−0.81, −0.18]	Higher porosity gives a smoother, less sharply peaked current. Magnitude unaffected.

Figure 1 shows the strongest relationship to porosity out of the the three data sets. The trend holds across the full porosity range rather than being driven by any outliers, and the error bars, determined from each electrode’s repeated measurements, are smaller than the markers, so the scatter about the

line is genuine variation between electrodes rather than measurement uncertainty. Every 95% C.I. stated is roughly 0.6 to 0.8 wide, which is why 20 electrodes establish plausibility in the correlations rather than predictive capability. Two further points support the results in the table. The broad optical background, covering 248 of the 256 spectral bins, shows no relationship with porosity at $r = +0.01$, ruling out an instrumental cause that would scale the whole spectrum. Additionally, the electrical result is uncorrelated with electrode thickness, so it is not a geometric measurement in disguise.

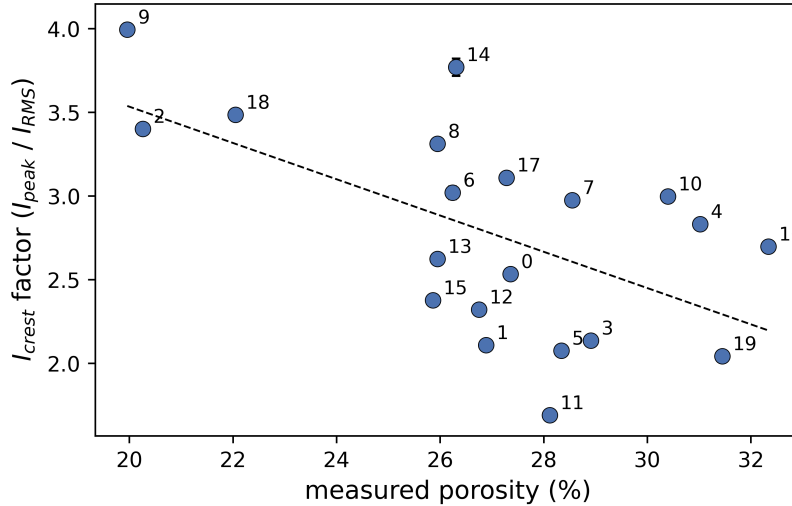


Figure 1: Current crest factor against porosity. Error bars show the standard error across each electrode’s 100 repeats and are smaller than the markers.

Three physically independent measurements agreeing in direction is stronger evidence than any single correlation, and is the principal reason we recommend continuing. One explanation consistent with all three is that a more porous electrode likely presents a greater accessible internal surface to the plasma, offered as a hypothesis for your specialists to assess.

3. Estimation Modeling & Accuracy

The measurements were reduced to 20 named physical quantities and a regularized linear model was fitted. More flexible methods were avoided, since with 20 electrodes they can reproduce the training data well, yet provide a lower accuracy prediction for new electrodes.

A R^2 parameter, representing the fraction of the variation in porosity that a model accounts for, was used to test the model. A R^2 value of 1 is perfect prediction, 0 is no better than always predicting the average porosity, and negative values are worse than that. Dividing the 2000 measurements at random places measurements of the same electrode in both training and test sets, so the model can recognize which electrode it is examining rather than learning a transferable relationship, and achieves $R^2 = 0.77$. Dividing instead by electrode reproduces production conditions, where an electrode arrives that has never been measured, and the same model achieves $R^2 = 0.21$. These model test results are described in the following table. Figure 2 makes the difference visible: predictions sit close to the diagonal in panel (a) but scatter widely in panel (b), although the model is identical in both. We report the lower electrode split model figure throughout, because it is the one that best describes performance on your line.

Approach	Inputs	R^2	Typical error (porosity % points)
Assume every electrode is average	0	-0.11	3.41
Use thickness alone	1	-0.11	3.41
Use all engineered measurements	20	+0.16	2.97
Use one summary per channel*	4	+0.22	2.85

* Selected after all 20 electrodes had been examined, so this is an optimistic upper bound rather than a validated result.

The best defensible model reduces typical error from 3.41 to 2.85 porosity percentage points, an

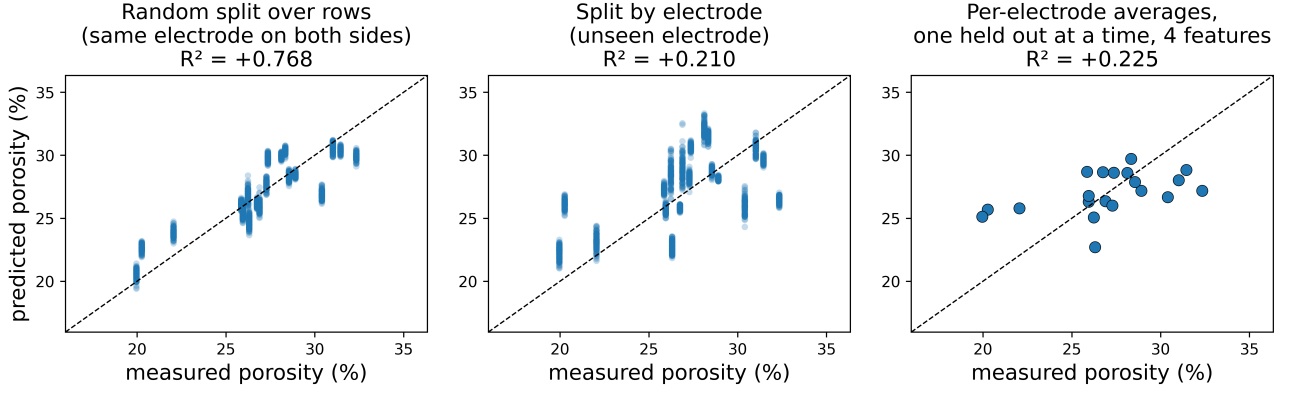


Figure 2: The same linear model assessed three ways. (a) Data divided at random. (b) Data divided by electrode. (c) Electrode averages, each held out in turn, using four quantities.

improvement of approximately 16%, on a property whose full range across the samples supplied is 12.4 percentage points. Given this minor reduction in error and R^2 , we do not recommend this model in place of offline characterization. Thickness alone provides no predictive value, so whatever the model achieves comes from the plasma characterization measurements.

4. Confidence and Required Study Size

The result is better than chance, but it is unstable: across the range of reasonable analytical choices the validated R^2 varies from -0.39 to $+0.31$, a spread wider than the effect being measured. This is characteristic of a study with too few samples rather than of an absent effect.

A power calculation indicates that confirming the strongest observed relationship would require approximately 21 electrodes if specified in advance, or 38 if a set of model parameter candidates is screened as performed in the present analysis. That same correlation of -0.58 carries 95% C.I. $[-0.81, -0.18]$ on 20 electrodes, narrowing to $[-0.75, -0.32]$ on 40 electrodes and $[-0.72, -0.38]$ on 60 electrodes. A larger study therefore delivers not only a porosity prediction for a new electrode, but a reliable one.

5. Recommendations for In-line Estimation

1. We recommend a characterization of 40 to 60 electrodes since our analysis shows that electrode count, not instrumentation, is the binding constraint.
2. We recommend deliberately spanning a wide porosity range, since a model can only be relied upon within the range on which it was established.
3. We recommend a standardized definition of the porosity prediction accuracy that would make an in-line measurement useful, so both parties can judge whether the follow-up study is worth undertaking.

6. Information Requested

- The method used to determine the reference porosity values, as these do not agree for the same electrode, and the answer determines which quantity is being estimated.
- Identification of the responding spectral peaks in terms of your process gas and electrode composition. These were selected by brightness alone, with no claim as to chemical origin.
- Confirmation of measurement units and calibration as the three datasets are supplied in arbitrary units, and approximately half of all optical intensity values are exactly zero, suggesting truncation at a noise threshold in the acquisition chain.